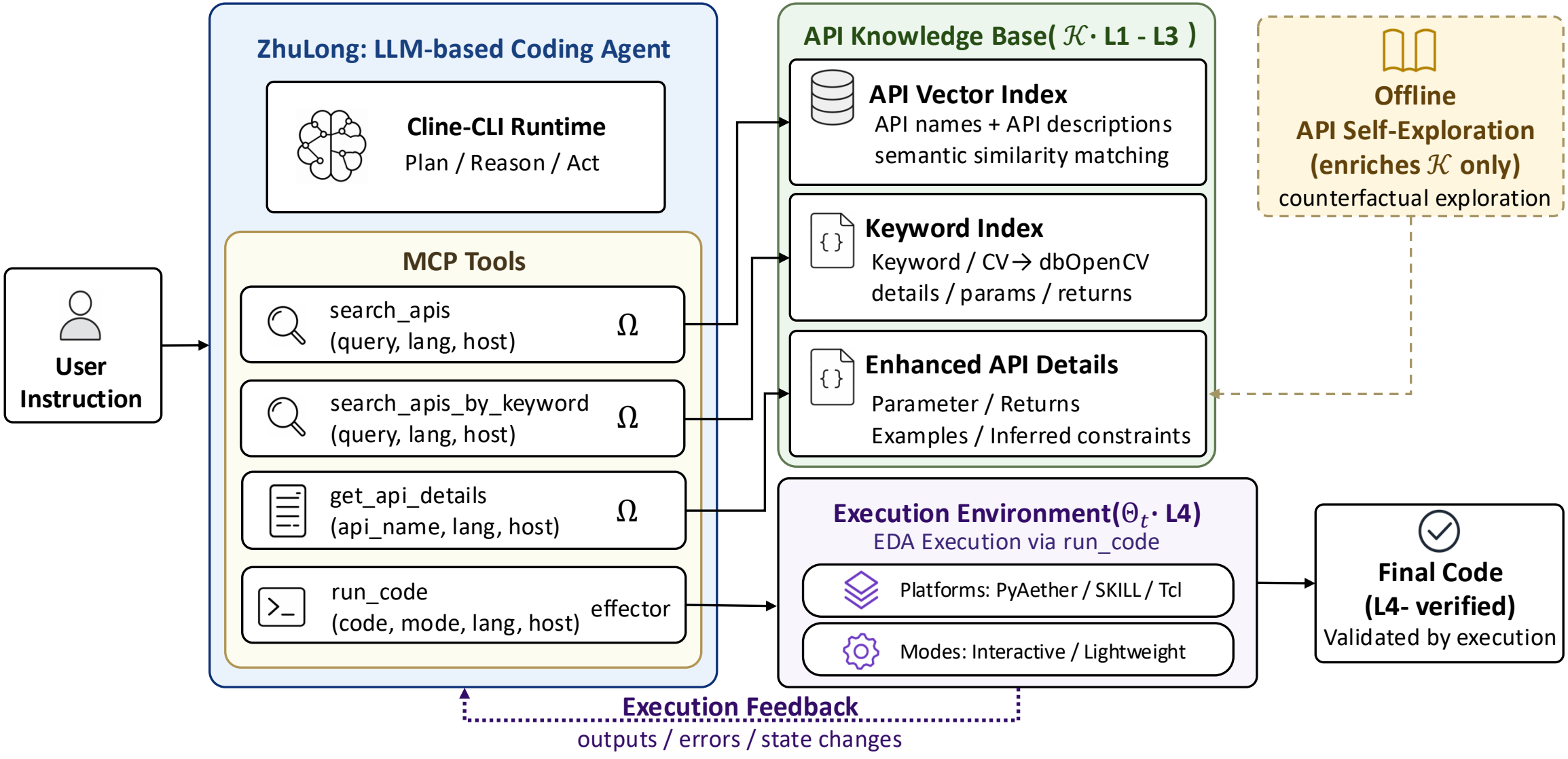

ZhuLong: LLM-based Coding Agent
Cline-CLI Runtime
Plan / Reason / Act
MCP Tools
search_apis
(query, lang, host)
Ω
search_apis_by_keyword
(query, lang, host)
Ω
get_api_details
(api_name, lang, host)
Ω
run_code
(code, mode, lang, host)
effector
User Instruction
API Knowledge Base(𝒦· L1 - L3)
API Vector Index
API names + API descriptions
semantic similarity matching
Keyword Index
Keyword / CV→ dbOpenCV
details / params / returns
Enhanced API Details
Parameter / Returns
Examples / Inferred constraints
Offline
API Self-Exploration
(enriches 𝒦 only)
counterfactual exploration
Execution Environment(Θ_t· L4)
EDA Execution via run_code
Platforms: PyAether / SKILL / Tcl
Modes: Interactive / Lightweight
Final Code
(L4- verified)
Validated by execution
Execution Feedback
outputs / errors / state changes